

DATA 23700

Assignment 1: Visualization design

See course schedule for the due date.

In this assignment, students will design a visualization for a small dataset, exchange critique with their peers, and produce a revised design informed by that feedback. The submission documents the whole process: the initial design and rationale, the critique the student found most useful, and the updated design.

In theory, students should be ready to explain the contribution of every pixel in the display; in practice, students are expected to explain choices about elements of the grammar of graphics such as marks, encodings, scales, labels, and annotations. Students may use any graphics or charting tool they please—including drafting a visualization by hand. However, students may find it most instructive to create the chart from scratch using a graphics API of their choice.

Students will *work alone* on their own designs. A peer critique session in class is a required part of the assignment.

Students should submit their assignment as a PDF document on Gradescope.

Antibiotics data

After World War I, antibiotics were considered as “wonder drugs”, since they were an easy remedy for what had been intractable ailments. To learn which drug worked most effectively for which bacterial infection, performance of the three most popular antibiotics on 16 bacteria were gathered. The numbers in the table represent the minimum inhibitor concentration (MIC) a measure of the effectiveness of the antibiotic. Specifically, MIC represents the concentration of antibiotic required to prevent growth in vitro. The reaction of the bacteria to Gram staining is described by the covariate “gram staining”. Bacteria that are stained dark blue or violet are Gram-positive. Otherwise, they are Gram-negative. The data is provided in the course repository at `data/antibiotics.csv`.

Technical specification

The assignment has three parts, submitted together as a single PDF.

Part 1: Initial design and rationale. Students will design a *static visualization* (i.e., a single image with no interactive elements such as tooltips) that they believe effectively communicates the antibiotics data, and they will accompany it with a short *write-up* (no more than 4 paragraphs) describing their design. While students must use the dataset provided, they are free to filter, transform, and augment the data as they see fit to highlight the elements they think are most important. **Students must use all columns in the provided dataset.**

Because different visualizations can emphasize different aspects of a dataset, students should document what aspects of the data they are attempting to most effectively communicate.

In short, what story (or stories) are they trying to tell? Just as important, students should note which aspects of the data might be obscured or downplayed due to their design choices.

Write-ups should provide a rigorous rationale for design decisions. Students should document the visual encodings used and why they are appropriate for the data. These decisions include the choice of mark type, encodings, scales, and other visual elements, as well as any use of sorting or other data transformations. How do these decisions facilitate effective communication?

Students should include a short description of the tools they used to create the visualization.

Part 2: Peer critique. Students will bring their initial design to class for a peer critique session. In small groups, students will exchange designs and offer constructive feedback in “I like” / “I wish” / “What if” form, considering choices about marks, encodings, scales, labels, annotations, and other elements of design. After the session, students will copy the three pieces of critique they found most actionable into their submission, verbatim.

Students who miss in the in-class critique session will need to gather design feedback from a peer outside of class in order to complete the assignment.

Part 3: Revised design and explanation of changes. Students will revise their initial design based on the critique they received. The revised image should appear in the submission alongside a short explanation (1–2 paragraphs) of what changed and why.

Submissions are evaluated as a whole on quality of visualization and quality of write-up.